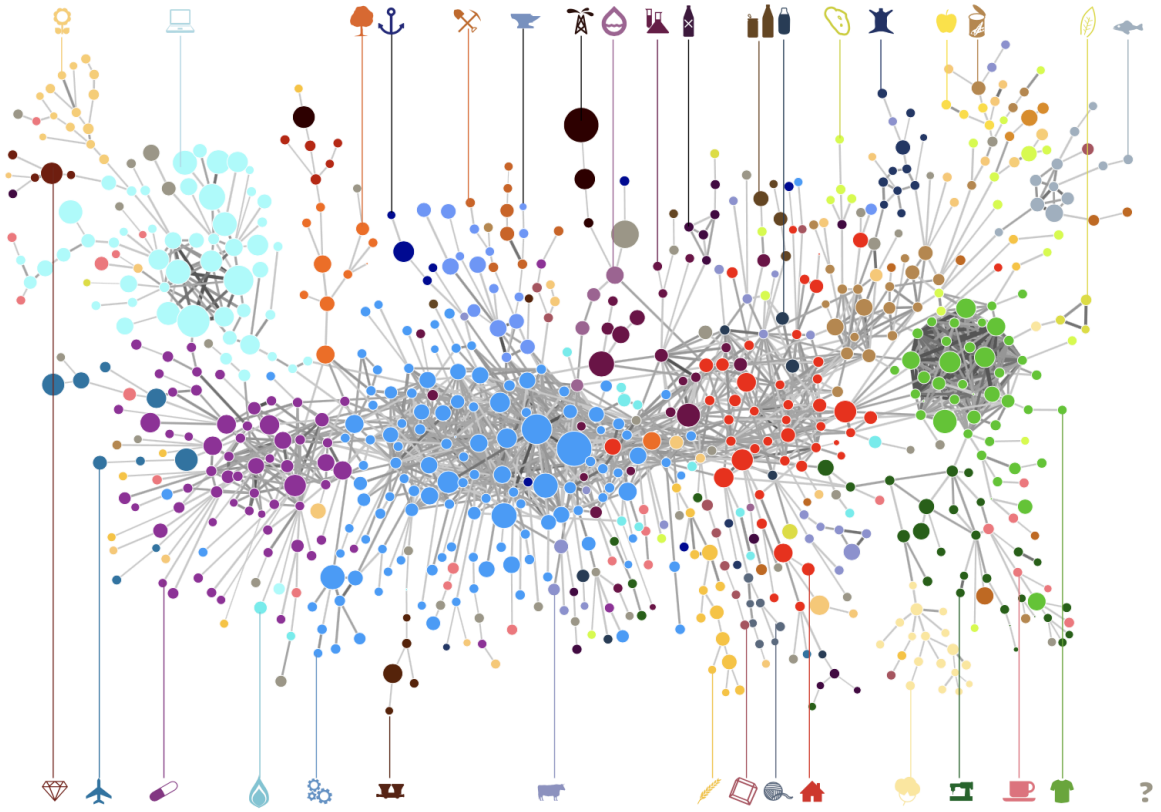


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Optimal Industry Development Strategy Using Economic and Product Complexity

United Kingdom

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The Economic Complexity Lab - Columbia University

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Optimal Industry Development Strategy for [Country]

0. Executive Summary

In this report, we aim to uncover the economic path the United Kingdom (UK) went through during the period from 2013 to 2023. Specifically, we dive into the topic through a macroeconomic overview of the UK from the perspective of Gross Domestic Product (GDP) and Economic Complexity Index (ECI), which reflect countries' production capability and production structure, respectively. We find that during the period of our interest, the UK demonstrated a high level of economic growth with an average annual GDP growth rate of 3.69% between 2019 and 2023, with a decreased economic complexity compared to other major developed economies.

The mixed progress during the period, we believe, can be partially explained through the analysis of the economic and product complexity structure of the UK on the industry group level, as well as the stagnant macro and microeconomic strategies. The product classification system we adopt allows us to measure the realized industry development for various industry groups and to observe that the UK progressed in developing most of its industries during the period while its developed counterparts and economies around the globe also adopted complex and progressive paths with variable degrees of success.

Based on our analysis of the opportunity gain and sensitivity analysis of major industry groups, the UK could further its gains by prioritizing Weapons. We further note that the UK has a heavily mismatched development strategy between 2013 and 2018 that indicates a low level of structural optimality (compared to other developed countries) on the complexity level, having an adjusted R-square of around 1.84%. Since a significant positive correlation exists between GDP growth and the structural optimality index of an economy as we define and calculate, the UK indeed has a suboptimal growth experience as compared with fast-growing economies both on a regional scale and on a global scale and has the potential to further its economic gains through industrial structure optimization as suggested in the paper.

The five years between 2019 and 2023 showed an improvement in the United Kingdom's economic growth, boosting its position on the global scale. Its GDP recovered to above the pre-pandemic levels, but it saw a decline in the ECI, particularly compared to its peer countries. The UK has the potential to further its economic gains through industrial structure optimization as suggested in the paper.

1. Report Overview

In this report, we aim to uncover the economic path the UK went through during the period from 2019 to 2023, before and after the COVID pandemic, compared to its performance from 2013 to 2018. Specifically, we dive into the topic through a macroeconomic overview of the UK from the perspective of Gross Domestic Product (GDP) and Economic Complexity Index (ECI), which reflect countries' production capability and production structure, respectively. We then analyze the industrial framework and changes of the UK during the period based on quantitative measures such as the Product Complexity Index (PCI) and Product Family Complexity Index (PfCI), followed by a detailed analysis of how such changes have matched with development strategies as quantitatively measured by opportunity gains on an industry-level. We conclude our report with macro-level results and predictions of the UK, as deduced from regression models we have established on a global scale, which take into account countries' stances in their product space.

2. Macroeconomic Results

2.1 GDP Ranking and Changes

Figure 1a illustrates the time-series GDP results for the UK and two of its peer countries (France and Italy) from 2013-2018. Similar to the other countries, the UK displayed upward growth over the period. Furthermore, the UK's average annual growth rate of 3.03% over the five-year period is comparable to those of its peers, which are 2.59% and 2.46% for France and Italy, respectively.

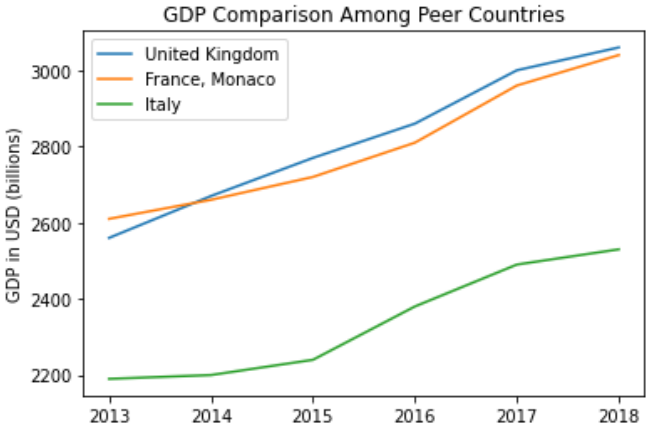


Figure 1a. GDP Comparison Among Peer Countries (2013-2018)

Figure 1b shows the GDP levels for the UK and its peer countries France and Italy from 2019 to 2023. Similar to the other countries, the UK displayed a drop in GDP from 2019 to 2020 because of the pandemic, with all GDPs rebounding in 2021. All three countries displayed a downward trend from 2021 to 2022, followed by a greater upward trend between 2022 and 2023. Over these five years, the UK's average annual GDP growth rate was 3.69%, which puts it higher than France's 2.49% and Italy's 3.05%.

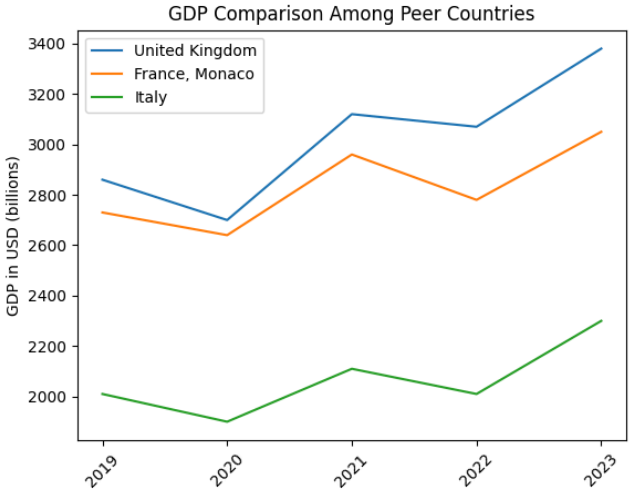


Figure 1b. GDP Comparison Among Peer Countries (2019-2023)

2.2 ECI Ranking and Changes

The Economic Complexity Index (ECI), as defined by Ricardo Hausmann and Cesar Hidalgo in 2009 to explain the knowledge accumulated in a country's population, is illustrated in Figures 2a and 2b.

As shown, the UK exhibits a significant increase in production capability from 2013 to 2018. The UK (63.22% ECI change) is the country in the peer group with the greatest increase in ECI change, as France attained a significantly lower ECI change (16.37%) and Italy (-31.46%) saw decreases in production capability over the period.

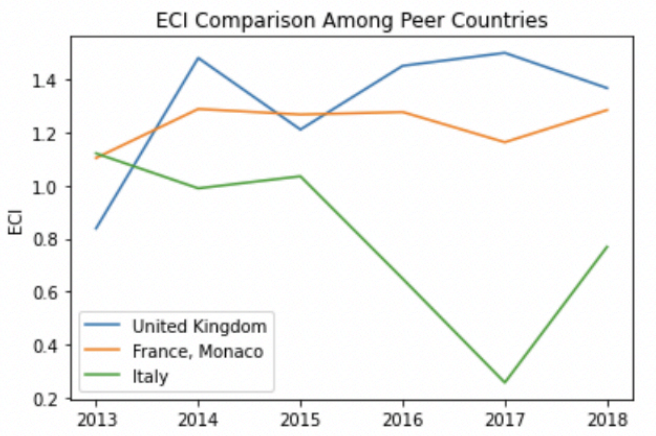


Figure 2a. ECI Comparison Among Peer Countries (2013-2018)

All three countries demonstrate a drastic decrease in ECI level between 2019 and 2020. The UK's ECI continued to decrease until 2022, while the comparable countries remained stable through 2021 and then began to increase. The UK is in a weaker position in terms of ECI, as its ECI change of -359.50% is much more extreme than France's -31.18% and Italy's -15.81%.

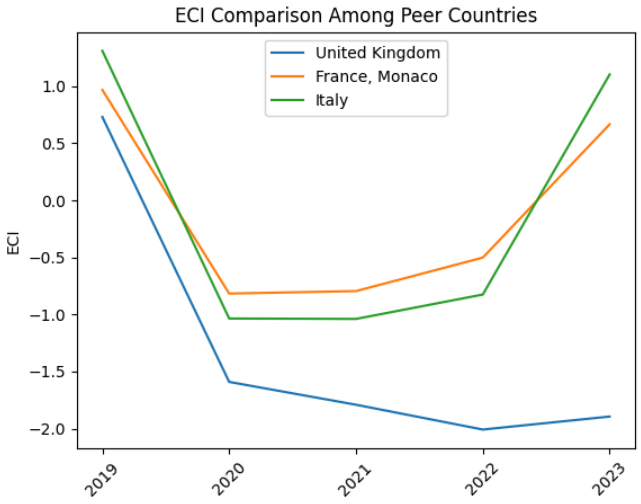


Figure 2b. ECI Comparison Among Peer Countries (2019-2023)

Figures 3a and 3b offer a further comparison of GDP Growth and ECI over the time span from 2013 to 2023 for the UK. The steady increase in GDP growth during the period from

2013 to 2018 is accompanied by a volatile eventual increase in ECI. The results show that the economic growth experienced by the UK during the period from 2013 to 2018, though relatively stable, is only partially consistent with the predictions by ECI and that the UK made some progress in effectively developing a competitive product space that helps to accumulate complicated knowledge and technology. For the period 2019 to 2023, the ECI and GDP both experienced downward trends. Then, between 2020 and 2023, the GDP demonstrated strong growth, while the ECI continued to decrease. In order to provide a more concrete answer to why this has happened, we will dive into an analysis of the UK's production structure and its product space designs in the next few chapters.

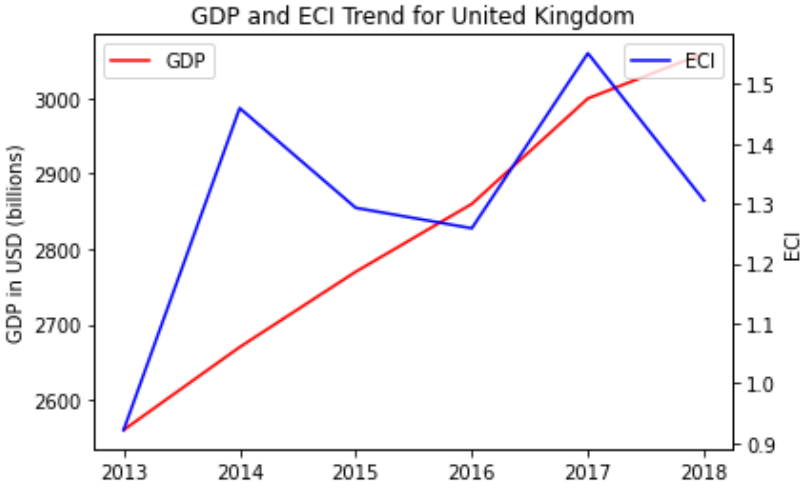


Figure 3a. GDP Growth and ECI for the UK (2013-2018)

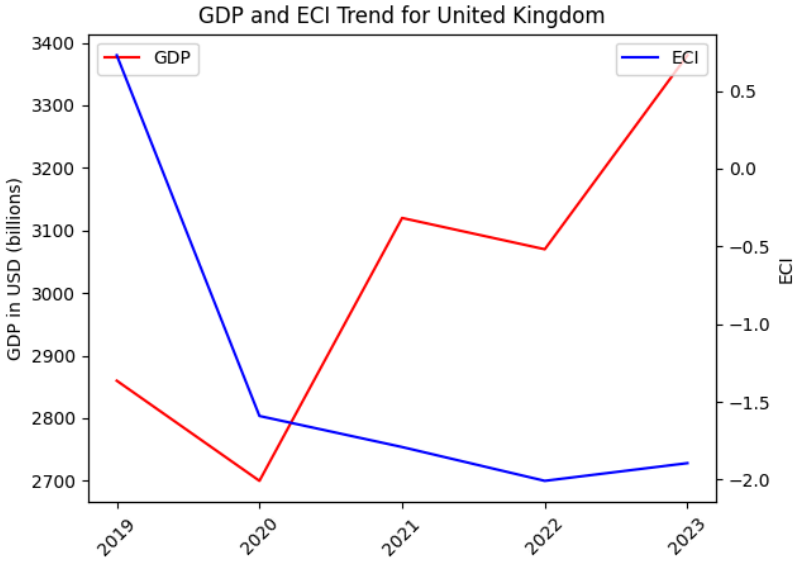


Figure 3b. GDP Growth and ECI for the UK (2019-2023)

3. Production Analysis

3.1 Product Categorization

Our main data source in this research is the UN Comtrade Website (https://comtrade.un.org), which provides us with valuable information on countries' export values on an annual basis, including highly specific export data for more than 5000 categories of products. In order to have a holistic vision of the product space of a country, however, a level of categorization is required. Products in our data source are labeled from 010000-999999, with the first two digits representing a general category to which the product belongs. Based on the 99 categories extracted from the label, we further group the products into the 18 most widely-accepted industries, as summarized in Table 1.

Note that the High-complexity Products industry group is not a distinct industry but rather an aggregated industry that encompasses the most complicated products from the most technologically-advanced industries (as measured by PCI). This industry serves as an indicator for us to quantitatively evaluate how well a country has performed in developing cutting-edge technologies and closing its technological gap with advanced economies, which has become a topic of wide concern today. Indeed, when we rank the 18 categories based on PfCI (an equivalent concept of PCI as calculated based on the 18 categorized product groups), the High-complexity Products industry has the highest PfCI among all industry groups.

Category	Description
Animals and By-products	All kinds of meat, livestock and related by-products
Vegetables, Fruits and Plants	All kinds of plants, edible vegetables and fruits
Food and Preparations	Edible preparations, raw cooking materials and appliances
Clothing and Accessories	All kinds of fabrics, filament, clothing and accessories
Textiles	All kinds of textile articles and related products
Wood Products	Book, paper and other elementary products related to wood
Metals	Mining of all sorts of raw Metals
Refined and Processed Stones	Refined or processed stone products such as pearl and ceramics
Energy Drilling and Mining	Mining industry for energy, minerals, stone materials, etc.
Daily Instruments	Basic daily instruments such as sport requisites, timing and measuring tools
Raw Manufacturing	Basic products made from plastic, rubber, leather, etc.
Chemical Manufacturing	Advanced chemical products and pharmaceutical products
Chemicals	Raw chemical materials and by-products
Machinery and Construction	Machinery and mechanical appliances, including electrical equipment
Transportation	Vehicles and locomotives including aircraft, ship, railway, etc.
Weapons	Explosives, arms and ammunition
Services and Utilities	Financial, business, legal, communication, travel Services and Utilities, etc.
High-complexity Products	Agglomeration of advanced products with highest PCIs

Table 1. Categorization of Products

We further observe a positive correlation between the GDP and the export of High-complexity Products. A higher adjusted R-square exists between GDP and High-complexity Products exports, indicating that more successful economics, as measured by GDP, tend to be those that are capable of producing and exporting High-complexity Products as we define them. Tables 2a and 2b show the OLS results in a universe of 107 and 21 typical countries respectively.

	High-complexity Products	Food and Preparations	Animals and By-products
Adj. R-squared	0.075	0.006	0.002
F-statistic	9.556	1.638	1.175
No. Observations	107	107	107

Table 2a. OLS Regression Results of GDP against Percentage Volume of Exports

	High-complexity Products	Food and Preparations	Animals and By-products
Adj. R-squared	0.149	0.003	-0.012
F-statistic	4.510	1.059	0.7542
No. Observations	21	21	21

Table 2b. OLS Regression Results of GDP against Volume of Percentage Exports

3.2 Realized Industry Development

Based on the aforementioned product categorization, we aim to capture the UK's realized industry production strategy between 2013 and 2023 on a complexity level. In Hausmann and Hidalgo (2009), distance is defined to be a measure of how far a country is from a certain product, not in its product space. We extend this concept to all product groups and quantify how technically equipped a country is to develop the corresponding product groups. In other words, the shorter the distance between a country and a product group, the more capable a country is of developing products within that group. Defining this net distance change as the Realized Industry Development (RID), our export data allow us to calculate and compare the RID of [Country] and the 18 product groups in 2018 and 2023 through the difference, where a positive RID indicates a shortening of the distance (i.e., advancement of the industry), and a negative RID indicates the reverse.

Category	RID
Transportation	-0.07617
Services and Utilities	-0.05456
Textile	-0.03133
Energy Drilling and Mining	-0.01078
Food and Preparations	-0.00908
Animals and By-products	-0.00673
Weapons	-0.00214
Raw Manufacturing	-0.00198
Vegetables, Fruits and Plants	-0.00047
High-complexity Products	0.00155
Metals	0.00367
Wood Products	0.00956
Precious Stones	0.01023
Clothing and Accessories	0.01242
Chemical Manufacturing	0.01394
Machinery and Construction	0.02269
Daily Instruments	0.03238
Chemicals	0.04009

Table 3a. Realized Industry Development (RID) of UK (2013-2018)

Category	RID
Clothing and Accessories	-0.00106
Wood Products	0.00638
Textile	0.01455
Vegetables, Fruits and Plants	0.01540
Services and Utilities	0.01902
Food and Preparations	0.01963
Animals and By-products	0.01994
Machinery and Construction	0.02095
Energy Drilling and Mining	0.02749
Metals	0.03571
Raw Manufacturing	0.03784
Weapons	0.04553
Chemicals	0.05594
Chemical Manufacturing	0.06166
Daily Instruments	0.06460
Precious Stones	0.06592
Transportation	0.08398
High-complexity Products	0.09677

Table 3b. Realized Industry Development (RID) of UK (2019-2023)

Between 2013 and 2018, the UK shows advancement in around half of the product groups, with negative RIDs pertaining mostly to low-complexity industries. Then, between 2019 and 2023, the UK demonstrates a progressive trend in ECI between 2019 and 2023, as almost all RID industries are positive.

3.3 Regional and Global Comparison

Tables 4a and 4b below summarize how the production strategy adopted by the UK compares with selected leading economies in the world across different years, as measured in RID. In 2018, France and Germany focused on high-complexity products. In 2023, leading economies such as France, Greece, and Portugal gave a considerable level of attention to wood products, energy drilling and mining, and precious stones.

Ranking	France	UK	Germany
1	High-complexity Products	Services and Utilities	High-complexity Products
2	Chemical Manufacturing	Weapons	Transportation
3	Daily Instruments	Raw Manufacturing	Wood Products
4	Animals and By-products	Precious Stones	Metals
5	Food and Preparations	Animals and By-products	Food and Preparations

Table 4a. RID Comparison across Leading Economies (2018)

Ranking	France	Greece	Portugal
1	Services and Utilities	Wood Products	Wood Products
2	High-complexity Products	Energy Drilling and Mining	Animals & By-products
3	Precious Stones	Clothing & Accessories	Textile
4	Transportation	Vegetables, Fruits and Plants	Vegetables, Fruits and Plants
5	Energy Drilling and Mining	Precious Stones	Clothing & Accessories

Table 4b. RID Comparison across Leading Economies (2023)

Tables 4c and 4d further examine the UK's development strategy against those of emerging economies on a global scale. Reasonably, emerging economies such as China and Singapore

have devoted the majority of their efforts to intermediate industry groups, showing slight shifts in strategies over the course of three years.

Ranking	China	Singapore
1	High-complexity Products	Transportation
2	Transportation	High-complexity Products
3	Machinery and Construction	Precious Stones
4	Energy Drilling and Mining	Daily Instruments
5	Daily Instruments	Services and Utilities

Table 4c: RID Comparison across Emerging Economies (2018)

Ranking	China	Singapore
1	Daily Instruments	Transportation
2	Chemicals	Weapons
3	Raw Manufacturing	High-complexity Products
4	Metals	Energy Drilling and Mining
5	Machinery & Construction	Machinery and Construction

Table 4d. RID Comparison across Emerging Economies (2023)

4. Product Space Design

4.1 Suggested Development Vector

As outlined above, distance gives us an idea of how far each new product is from a country's current mix of exports. As defined in Hausmann and Hidalgo (2009), distance is "the weighted proportion of products connected to goods p that country c is not exporting, i.e., MCP = 0." Based on this definition, the distance between a country and any product that the country is currently exporting (i.e., the country is a significant exporter of the product as compared to the global average) is zero. Similarly, opportunity gain quantifies the contribution of a new product in terms of opening up the doors to more and more complex products. Intuitively, a product that a country is currently exporting (i.e., MCP = 1) has an opportunity gain of zero. An unreasonable investment in a certain product p would lead to a relative decrement in the export of more profitable products and therefore lead to a negative opportunity gain of product p. The opportunity gain is not the only estimator of a country's level of economic development. However, among countries with comparable economic strength, those having higher ECI, or in other words, more reasonable economic structures are often in better shape.

Ranking	United Kingdom	France	Italy	Greece	China	Singapore
1	High-complexity Products	Energy Drilling and Mining	Precious Stones	Precious Stones	Energy Drilling and Mining	Machinery and Construction
2	Transportation	Precious Stones	Energy Drilling and Mining	Clothing and Accessories	Textile	Chemical Manufacturing
3	Machinery and Construction	Textile	Vegetables, Fruits and Plants	Weapons	Chemical Manufacturing	Energy Drilling and Mining
4	Services and Utilities	Wood Products	Food and Preparations	Services and Utilities	Vegetables, Fruits and Plants	Chemicals
5	Chemicals	Animals and By-products	Chemical Manufacturing	Chemicals	Chemicals	Transportation
6	Daily Instruments	Vegetables, Fruits and Plants	Raw Manufacturing	Daily Instruments	Food and Preparations	Daily Instruments
7	Chemical Manufacturing	Food and Preparations	Clothing and Accessories	Raw Manufacturing	Raw Manufacturing	Services and Utilities
8	Raw Manufacturing	Chemicals	Textile	Transportation	Clothing and Accessories	Weapons
9	Weapons	Chemical Manufacturing	Metals	High-complexity Products	Transportation	Wood Products
10	Precious Stones	Raw Manufacturing	Machinery and Construction	Animals and By-products	Weapons	Metals
11	Wood Products	Clothing and Accessories	Daily Instruments	Vegetables, Fruits and Plants	Daily Instruments	High-complexity Products
12	Metals	Metals	High-complexity Products	Food and Preparations	Precious Stones	Raw Manufacturing
13	Clothing and Accessories	Weapons	Animals and By-products	Energy Drilling and Mining	Services and Utilities	Textile
14	Textile	High-complexity Products	Weapons	Chemical Manufacturing	Wood Products	Precious Stones
15	Food and Preparations	Machinery and Construction	Wood Products	Wood Products	Animals and By-products	Clothing and Accessories
16	Animals and By-products	Daily Instruments	Chemicals	Textile	Metals	Animals and By-products
17	Energy Drilling and Mining	Services and Utilities	Services and Utilities	Metals	Machinery and Construction	Food and Preparations
18	Vegetables, Fruits and Plants	Transportation	Transportation	Machinery and Construction	High-complexity Products	Vegetables, Fruits and Plants

Table 5. Opportunity Gain Ranking on a Comparable Scale (2023)

We define the ranking of opportunity gain for a country c, as the Suggested Development Vector (SDV) of c. Table 5 summarizes how the SDV of the UK compares with a selected group of comparable and leading economies in the world, with the industry groups ranked in decreasing order of significance.

Dark blue indicates positive opportunity gains, light blue indicates zero opportunity gains, and white indicates negative opportunity gains. The result shows that leading countries barely have any industries with positive opportunity gain. In other words, those countries are already in the most stable and efficient economic structure. Without the necessity of economic transition, those countries merely need to maintain a steady and balanced economic development. In contrast, developing countries have nearly half of industries with positive opportunity gain and share common interests in Energy Drilling and Mining and Textile industries. It can be concluded that the United Kingdom is categorized to be a highly complex country as compared with the world's leading economies and other major economies in Europe.

Ranking	Thailand	Italy	Spain	United Arab Emirates
1	Animals and By-products	Energy Drilling and Mining	Energy Drilling and Mining	High-complexity Products
2	Food and Preparations	Precious Stones	Precious Stones	Transportation
3	Chemicals	Vegetables, Fruits and Plants	Weapons	Machinery and Construction
4	Raw Manufacturing	Animals and By-products	Textile	Daily Instruments
5	Textile	Food and Preparations	Chemicals	Chemicals
6	Machinery and Construction	Chemical Manufacturing	Daily Instruments	Weapons
7	Transportation	Raw Manufacturing	Animals and By-products	Textile
8	High-complexity Products	Wood Products	Vegetables, Fruits and Plants	Chemical Manufacturing
9	Energy Drilling and Mining	Clothing and Accessories	Food and Preparations	Raw Manufacturing
10	Vegetables, Fruits and Plants	Textile	Chemical Manufacturing	Energy Drilling and Mining
11	Precious Stones	Metals	Raw Manufacturing	Precious Stones
12	Weapons	Machinery and Construction	Wood Products	Services and Utilities
13	Clothing and Accessories	Daily Instruments	Clothing and Accessories	Metals
14	Services and Utilities	High-complexity Products	Metals	Clothing and Accessories
15	Wood Products	Weapons	Transportation	Wood Products
16	Metals	Chemicals	Services and Utilities	Animals and By-products
17	Chemical Manufacturing	Services and Utilities	Machinery and Construction	Vegetables, Fruits and Plants
18	Daily Instruments	Transportation	High-complexity Products	Food and Preparations
ECI	1.68914409	1.179705915	0.45830144	-0.734459148

Table 6. SDV of Other Sample Countries (2019)

4.2 Budgeted Industry Development Strategy and Interpretation

The optimal budgeted industry development strategy aims to solve the most efficient allocations under the budgeted constraint of different situations of heuristic total export growth. In order to obtain such an allocation portfolio, we formulate an optimization problem as the following:

$$\max_{x \in \mathbf{R}^{|P|}} \quad \left(\frac{k(x) - < k(x) >}{stdev(k(x))}\right)_C$$

Subject to

$$\sum_{p \in P} x_p \leq B$$

$$x \succcurlyeq 0$$

where k(x) is the second largest eigenvector of $M_{CC'}$ after adding x to the original export vector, C is the country that we are interested in, P is the product space, B is the heuristic budget.

If we can solve this problem, then we can know what the largest ECI to which a country can improve. Accordingly, the optimal solution x * that we obtain here is exactly our optimal allocation. However, the biggest problem is that the objective function is not convex. Therefore, we cannot apply the classical tools for convex optimization to this problem. In order to solve this problem, our idea is to use a greedy algorithm. The way that we apply the greedy algorithm is that firstly we separate the total budget into k equal parts, and then for the first equal part we try to search for the product that can optimize our objective function by iterating through each $p \in P$. After we find the optimal product p for this equal part, we add the amount back to the export value of this product p and update the ECI for this country. If we observe that ECI does not change after iterating all the products, then we can add up one more equal part to invest. Until we find there is an ECI increment, we add the amount now to that product. The equal rest parts can be done in the same way.

By definition, a greedy algorithm is a kind of local search algorithm that is possible to run into a local minimum. Therefore, for local search algorithms, one typical way to keep it away from the local minimum is to enhance the vision of the algorithm by looking more steps forward. In our setting, whenever we obtain an increment of ECI, we mark down the required amount of export growth. Simultaneously, we continue searching for the next ECI increment point, compare the ratios of these two possibilities and pick the largest one. The ratio is defined as the following:

$$Ratio(p) = \frac{ECI_p - ECI_0}{||x_p - x_0||_1}$$

where ECI_p is just the ECI of the country after adding some amount of export increment to product p, which is exactly the amount that is able to make ECI change. Also, we can express it in the way of $||x_p - x_0||_1$.

In this way, we can obtain a further sighted greedy algorithm. Also, the number of parts that we would like to separate the budget into also matters. In order to solve this problem, we added a stabilizer on top of the enhanced greedy algorithm by running the algorithm for different numbers of parts and selecting the portfolio with the largest ECI increment and most complex economic structure.

Besides, it is quite common that there is a surplus in the budget. To deal with the surplus, we just add it to the product which needs the least amount of dollars to make $M_{cp} = 1$ (i.e., the smallest amount of required export increase).

Economic Panel

As stated above, the enhanced greedy algorithm is developed mainly for solving the optimal allocation strategy. In order to visualize this allocation, an economic panel is created, consisting of a pie chart, a line chart and a table. First, a pie chart is used to show the percentage of each product in the optimal portfolio. The number located in the center of the circle denotes the heuristic growth of total export, which is also our budget. Each category in the chart has a determined percentage of the allocated budget. The lines below the pie charts show how the ECI changes during the searching process of the greedy algorithm. Even though sometimes an increase of the export by 15% or 20% does not seem reasonable to developed countries, yet it can also give a sense of how a country should develop in the long term. Eventually, the graph in the end is just for numerical explanation for the pie charts.

Interpretation

For the purpose of a clear explanation of the line charts, we label all the products in Table 7. Please refer to this table when trying to understand the investment trajectories in the line charts.

Label	Product Names
1	Animal and By-products
2	Vegetables, Fruits and Plants
3	Food and Preparations
4	Energy Drilling and Mining

5	Chemicals
6	Chemical Manufacturing
7	Raw Manufacturing
8	Wood Products
9	Clothing and Accessories
10	Textile
11	Metals
12	Machinery and Construction
13	Transportation
14	Weapons
15	Daily Instruments
16	Precious Stones
17	High-complexity Products
18	Services and Utilities

Table 7. Products Labeling

Figure 4 shows an economic panel for the UK in 2023. We can see that for different percentages of heuristic total export growth, the ECIs of the United Kingdom actually have similar amounts of increments.

For 2023, a 5 percent increase in export would emphasize Services and Utilities, followed by Services and Utilities, Weapons, and Weapons for 10, 15, and 20 percent export growth respectively. These strategies go in line with the suggested trajectories based on ECI. The change from a 5 to 10 percent increase in export decreased the focus on Services and Utilities and Chemicals, while increasing the focus on Daily Instruments and Weapons. Both the 15 and 20 percent optimizations increased exports for Weapons and reduced exports in other sectors evenly. A more detailed numerical representation of the portfolio for the United Kingdom is shown in Table 8.

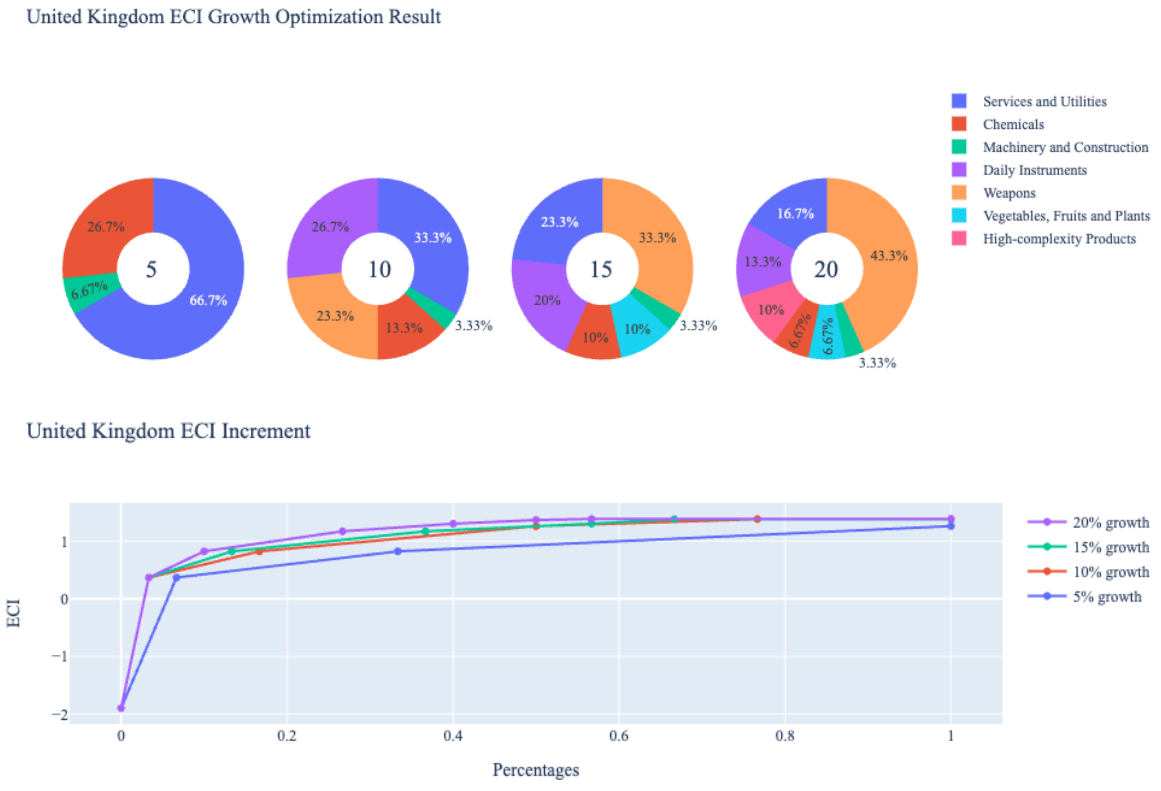


Figure 4. Greedy-Based ECI Improvement Results of UK with Different Budgets (2023)

Heuristic Export Growth	5%	10%	15%	20%
Vegetables, Fruits and Plants	0	0	3308527171.975999	2940913041.755999
Chemicals	2940913041.757333	2940913041.756000	3308527171.976000	2940913041.756001
Machinery and Construction	735228260.439333	735228260.439000	1102842390.658667	1470456520.878000
Weapons	0	5146597823.073001	11028423906.586664	19115934771.413998
Daily Instruments	0	5881826083.511998	6617054343.952002	5881826083.512001
High-complexity Products	0	0	0	4411369562.633999
Services and Utilities	7352282604.393334	7352282604.390000	7719896734.610667	7352282604.389999
Increased ECI	1.259068	1.380261	1.378668	1.384459

Table 8. Portfolio of UK (2023)

4.3 Sensitivity Analysis and Contingency Strategy

In the following section, our broad aim is to understand how sensitive the UK is to changes in export in various industries. While the Economic Complexity Index has been shown to contain valuable information about the economic strength/capacity of a country, little has been said as to the stability of the measure itself. We seek to uncover the main drivers of stability under a country's ECI score and better understand longer-term projections for ECI growth. Additionally, gaining an intuition on the industries that are bolstering an economy and the industries that offer the greatest return per unit of investment is best done through visualization of industry-specific sensitivity. To this end, we have developed two categories of sensitivity analysis.

ECI Sensitivity Graph

First, in order to assess the effect particular industries have on each respective country, we developed ECI Sensitivity graphs (Figure 5). We use the shocking methodology to increment or decrement an industry’s export level by a certain percentage and recalculate the ECI, all else being equal. While freezing the export dynamics of the rest of the world and enacting an instantaneous shock in export for a particular industry and country may not reflect real-life dynamics, what it does offer us is an understanding of a country’s level of dependency on an industry in the context of ECI. The shocking is done in small percentage increments in order to smooth out the resulting curve. Additionally, a recurring theme one will see is that for most industries, a breaking point occurs where the ECI either jumps upwards or downwards, creating a hockey stick shape rather than a linear trend. While the inflection point may not be accurate in the context of real life, comparing the various inflection points across industries can give us valuable insight.

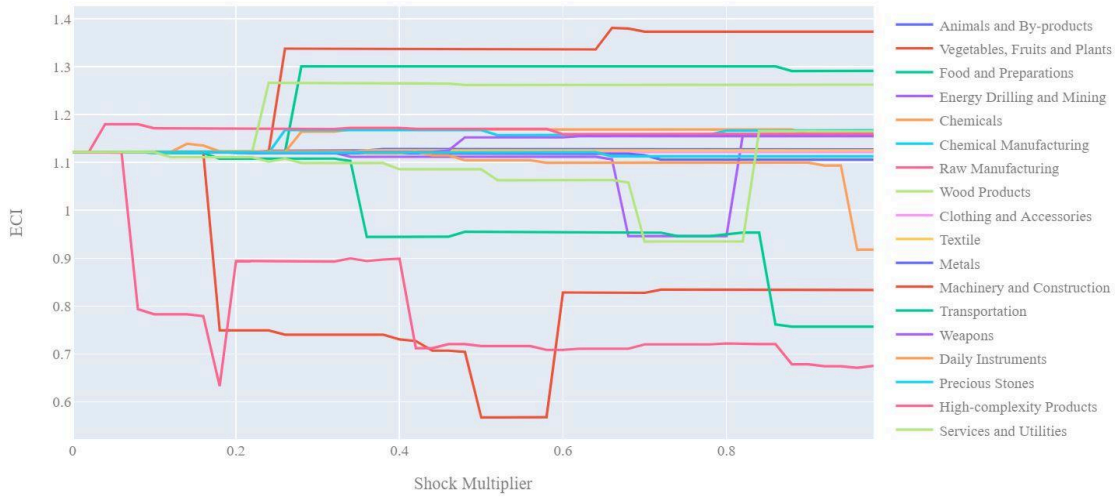


Figure 5. Sample ECI Sensitivity Graph (2018)

The graphs are formulated as follows. There are six ECI Sensitivity graphs, two for each of the following years: 2013, 2018, and 2023. The first chart, Incremental ECI Sensitivity, shows positive shocking in which export numbers are shocked upwards per industry. The X-axis is in terms of percentage shock per current dollar value of export. The Y-axis is the recalculated ECI level post-shocking. The second chart, Decremental ECI Sensitivity, follows the same format except the X-axis represents negative shock. In other words, exports for each industry are decremented, and the ECI is accordingly recomputed.

4.3 Sensitivity Analysis Results and Interpretation

We examine the UK's ECI Decremental and Incremental Sensitivity Charts for 2013 to 2023. Figure 6a shows that in the positive shocking case, even a small increase in High Complexity Products could drastically improve the ECI index. Similarly to 2018 in Figure 6c, an increase in Transportation would also increase ECI level. Increases in export of most other industries lead to decreasing ECI scores.

In the 2013 negative shocking case, we see that again the UK could benefit from a decrease in Transportation and Precious Stones. In addition, large decreases in Energy, Drilling, and Mining could increase the ECI score. In 2013, the UK was incredibly sensitive to reductions in Machinery and Construction.

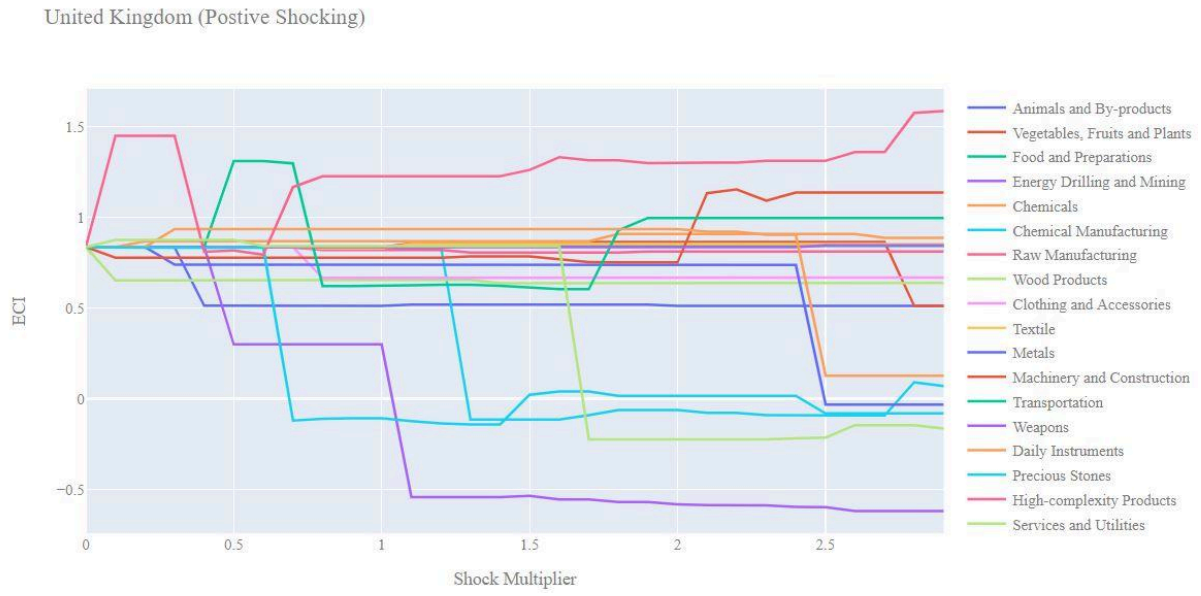


Figure 6a. Incremental ECI Sensitivity for UK (2013)

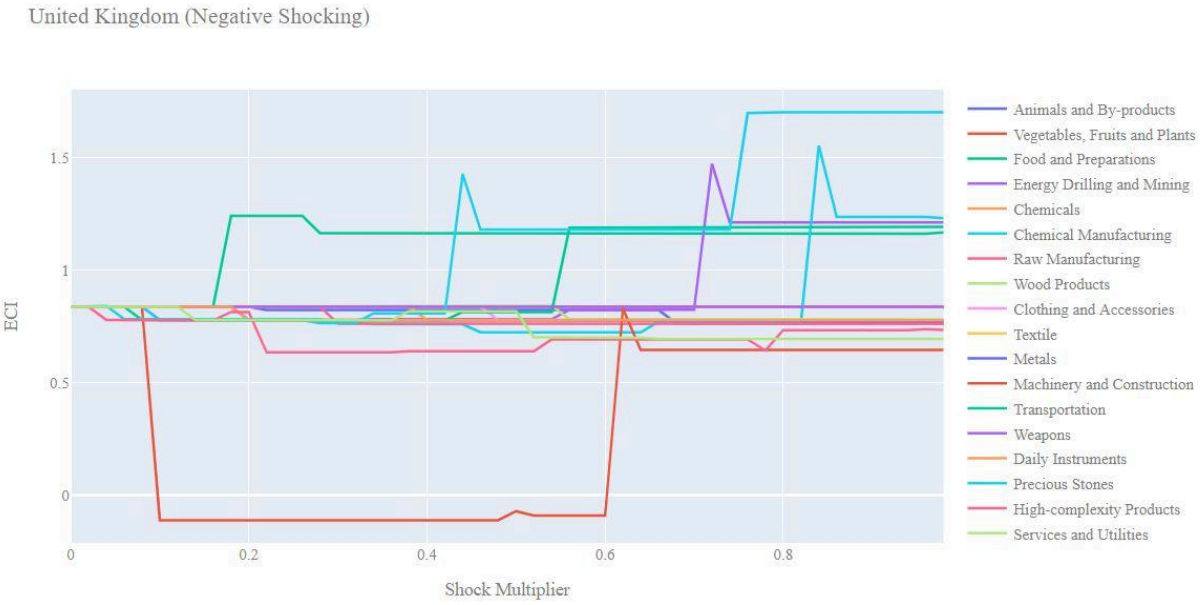


Figure 6b. Decremental ECI Sensitivity for UK (2013)

Below, Figures 6c and 6d show the United Kingdom's 2018 ECI Sensitivity Charts. Firstly in the positive chart, we notice immediately that an increase in export in the majority of industries actually causes decreases in ECI. Only Transportation seems to give an immediate increase in ECI level. A large increase in High Complexity Products would also lead to an increase in the ECI level.

Next we consider the United Kingdom's 2018 Negative Shock chart. We see that for the UK decreasing the export of Food and Preparation could drastically increase the overall ECI for the country. Similarly large decreases in Precious Stones could also increase ECI level. The

UK is most sensitive to decreases in export of High Complexity Products and Machinery and Construction.

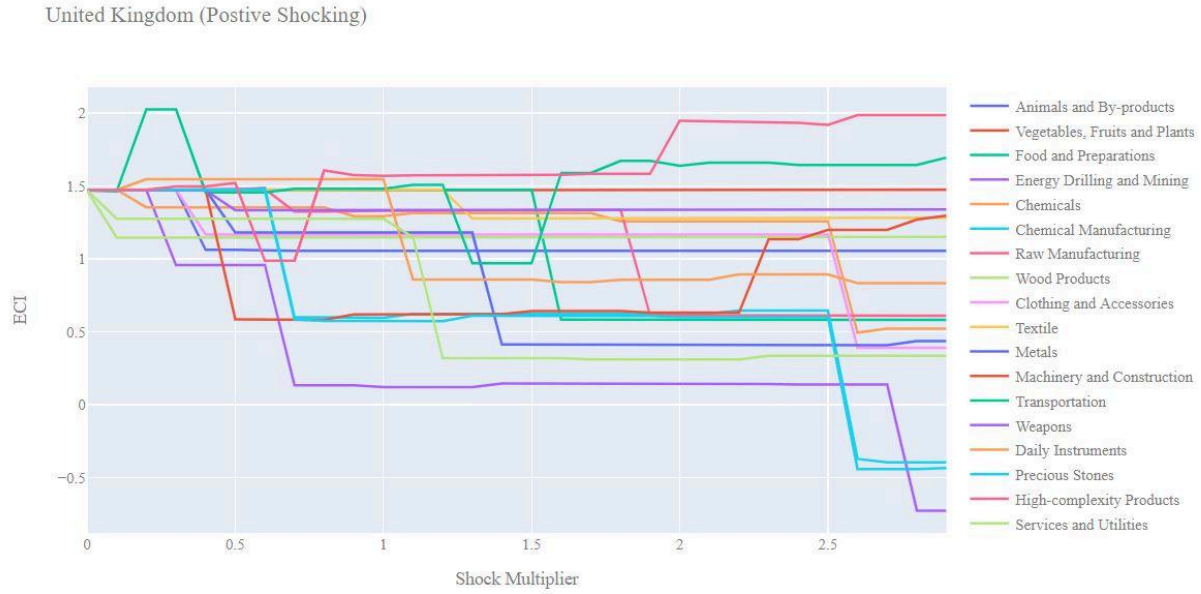


Figure 6c. Incremental ECI Sensitivity for UK (2018)

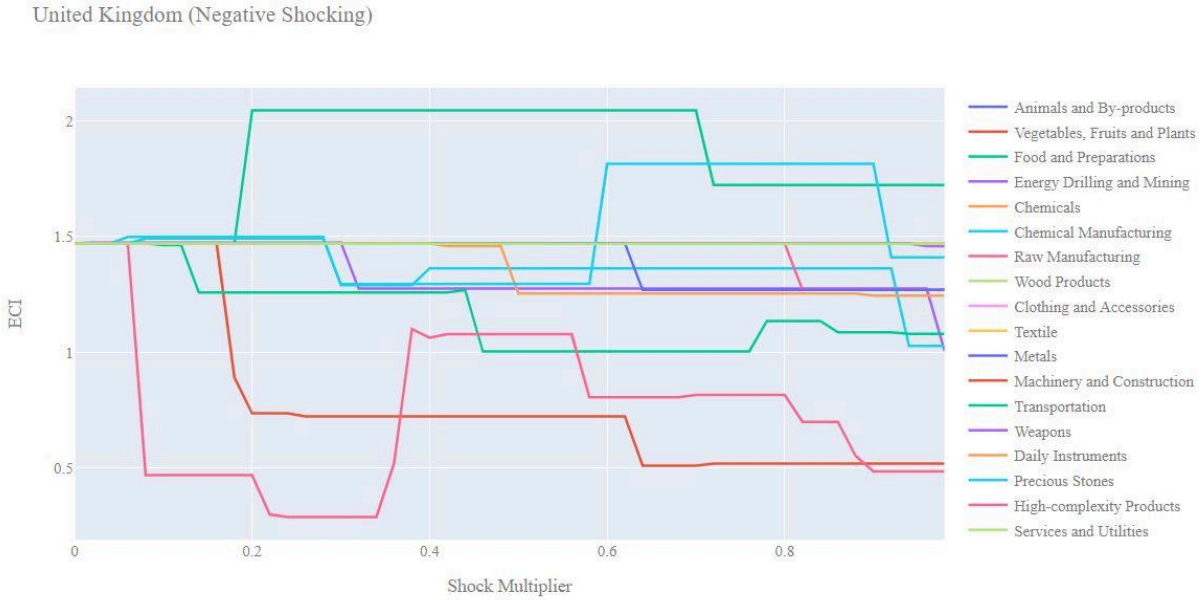


Figure 6d. Decremental ECI Sensitivity for UK (2018)

Figures 6e and 6f below demonstrate the UK's ECI Sensitivity Charts for 2023. Unlike prior years, there were no increases in ECI associated with raising or lowering exports in any industries. Many industries were represented with relatively flat lines, which demonstrates that changing the export volume of those sectors would have a negligible impact on the UK's ECI. Figure 6e demonstrates that increasing exports in

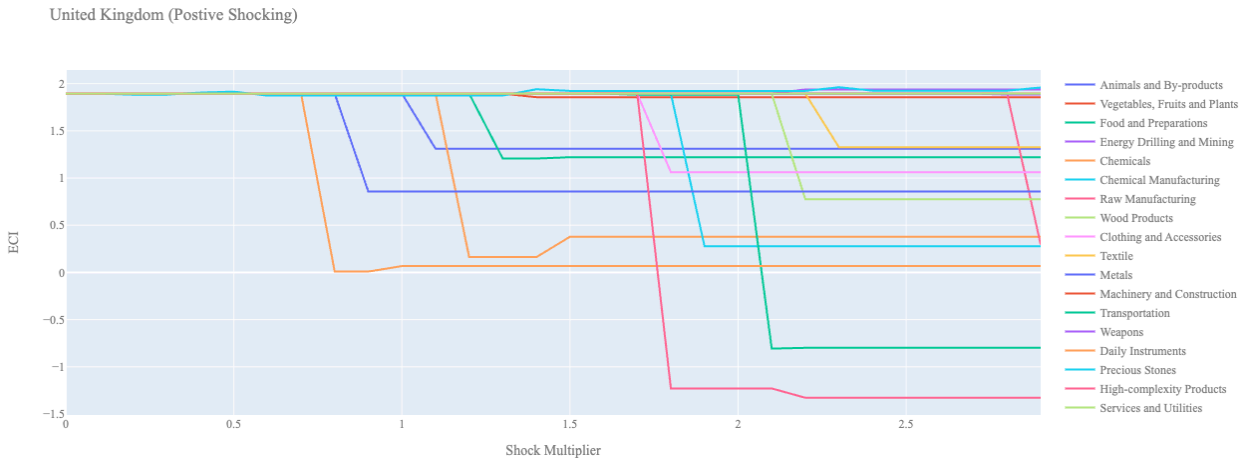


Figure 6e. Incremental ECI Sensitivity for UK (2023)

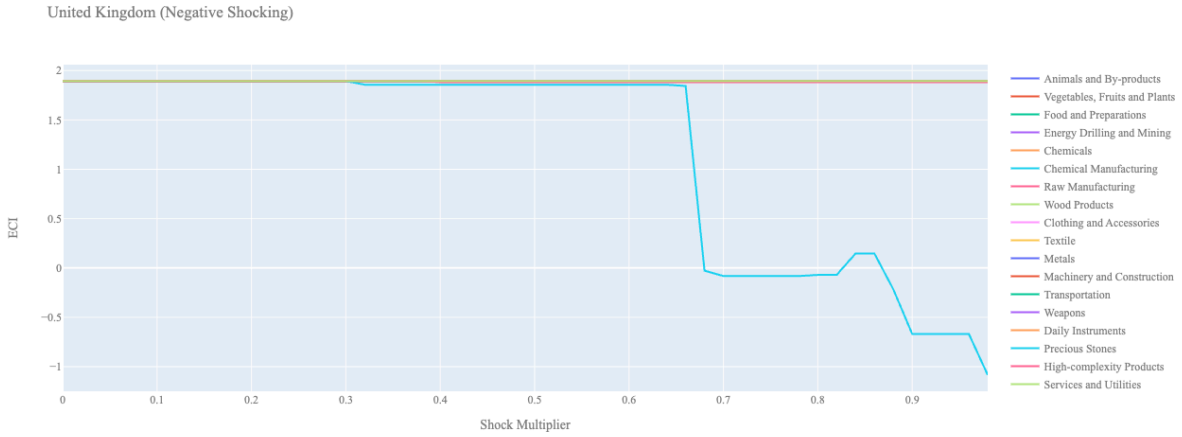


Figure 6f. Decremental ECI Sensitivity for UK (2023)

5. Macro-level Predictions

5.1 Structural Optimality Index

ECI sensitivity described in the section before has proven to be a significant factor in deciding the economic outcomes of countries. In order to quantitatively evaluate how effective our development strategies are and how deviations from the suggested developmental path will lead to suboptimal growth results, we define the measure – Structural Optimality Index (SOI) – on a country level as the Adjusted R-Square obtained from regressing the country's realized industry development over the suggested development strategy. Both strategies are quantified in nature, with the realized development represented by increasing export in every major industry group and the suggested development strategy represented by the opportunity gains of developing the corresponding industry groups as calculated in previous sections. The adjusted R-square from the regression thus provides us with a quantified measure of how the development path adopted by the country has fitted with/differed from calculations on a complexity level.

After obtaining this measure for all countries in our database, we further regress countries' five-year economic growth, which is defined as the percent increase in GDP (USD), over

their SOI. Figure 9 illustrates the regression output, which shows a positive correlation between countries' SOI and their five-year GDP growth, significant at the 1% level and achieving an adjusted R^2 of 4.5%. In other words, the more a country develops, as suggested by calculations on a complexity level, the better economic outcomes it achieves.

Table 14: OLS Regression Results	
Dep. Variable: GDP	**F-Statistic:** 5.963
R^2: 0.054	**Prob (F-Stat):** 0.0163
Adj R^2: 0.045	**No. Observations:** 106

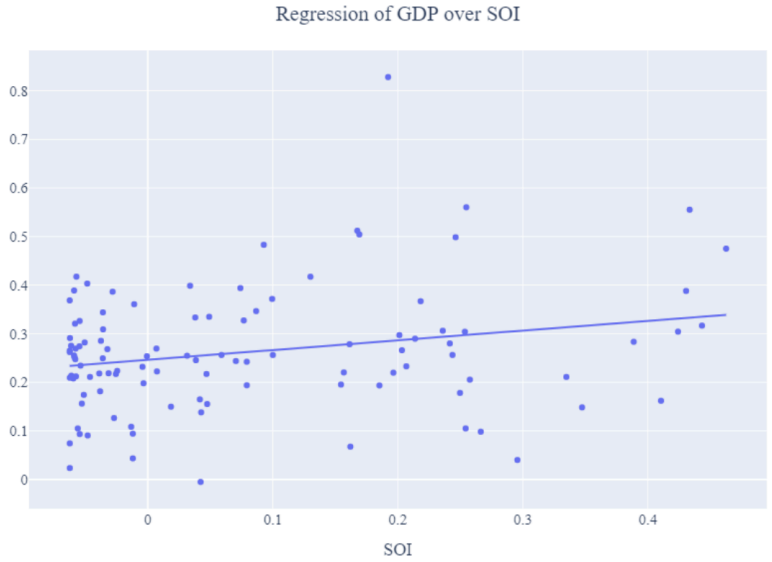


Figure 9. Regression of Five-Year GDP Growth over SOI

5.2 SOI of United Kingdom and Its Implications

The United Kingdom, for instance, has a low level of structural optimality as compared with its developed peers. As the regression in Figure 4 suggests (the orange curve represents export increase and the blue curve stands for opportunity gain of the corresponding industry), the UK demonstrates a low level of correlation with the suggested developmental path, having an adjusted R-square of around 1.84% in 2018. The U.K. ranks in the 4th quartile among all countries in terms of SOI and makes an average rate of economic progress lower than its developed peers from 2013 to 2018 with an average 5-year GDP growth rate of 3.03%, which is consistent with the regression result. This is in direct contrast to emerging economies such as Turkey, where there is high SOI of around 21.8% and a corresponding GDP growth of 36.7% from 2013 to 2018, making it one of the fastest developing economies during the period. In sum, the U.K.'s slow economic progress from 2013 to 2018 can be at least partially explained by a development strategy that demonstrates a low level of similarity with our suggested developmental path. In specific, the U.K. invested heavily in the Chemicals and Machinery industry, while it should have focused more on industry groups such as Energy & Drilling, and Precious Stones.

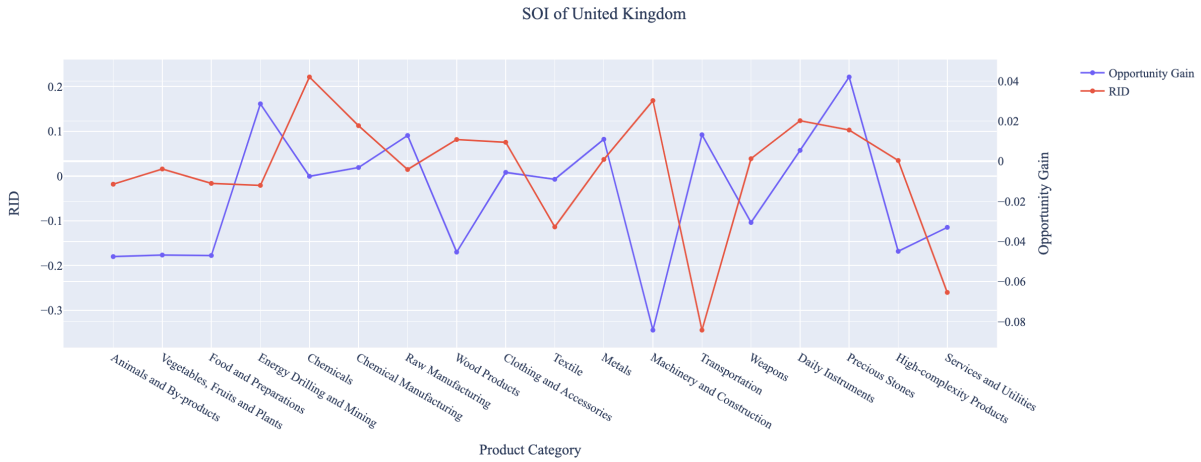


Figure 10a. SOI of UK (2013-2018)

In 2023, the UK ranks in the second quartile among all countries in terms of SOI and has made a fast rate of economic progress among its emerging peers from 2019 to 2023 with a 5-year GDP growth rate of 3.69%, which is consistent with the regression result.

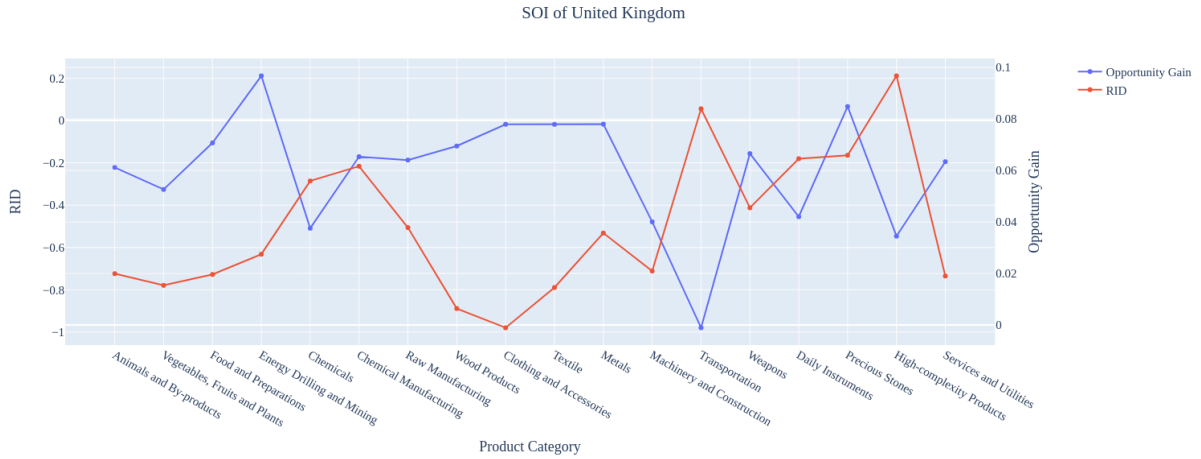


Figure 10b. SOI of UK (2019-2023)

6. Concluding Remarks

In conclusion, the UK demonstrated a low level of economic growth in terms of annual GDP increase during the period from 2013 to 2018, concurrent with an increase in economic complexity as compared with other major developed economies. For 2019-2023, the GDP growth has increased, alongside a decrease in ECI as compared with other developed countries.

We believe that the mixed progress between 2013 and 2018 can be partially explained through the analysis of the economic and product complexity structure of the UK on the industry group level. The product classification system we adopt allows us to measure the realized industry development for various industry groups and to observe that the UK consistently relied on developing the Chemicals, Machinery and Construction, and Daily Instruments industries during this period, while some of its developed counterparts and emerging economies around the globe adopted much more complex and progressive paths. Between 2021 and 2023, the mixed progress is a result of global demand for basic, low-complexity products, which the UK positions itself to meet. As such, its ECI decreases, but its GDP increases at a faster rate, in average, than its developed counterparts.

Based on our analysis of the opportunity gain and a sensitivity analysis of major industry groups, the UK could further its gains by prioritizing High-Complexity Products and Transportation. Given the economic effects of COVID-19, the UK's regression in terms of complexity leads us to suggest the UK further its gains by focusing on Weapons. We further note that the UK had a heavily mismatched development strategy in 2013-2018 that indicated a low level of structural optimality (compared to other developed countries) on the complexity level, having an adjusted R-square of around 1.84%. Since a positive correlation exists between five-year GDP growth and the structural optimality index of an economy as we define and calculate, the UK indeed has a suboptimal growth experience as compared with fast-growing economies both on a regional scale and on a global scale, and has the potential to further its economic gains through industrial structure optimization as suggested in the paper.

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